

David William Erwin

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EXPERIENCE

Software Engineer

Mastercard

September 2025 – Present

Prague, Czech Republic

- Built and shipped full-stack features for a unified platform with product and engineering teams, spanning React and TypeScript interfaces and .NET and C# services backed by Databricks.
- Co-led a two-engineer team building Tourism Insights, a new product that replaced two legacy pipelines, expanded their capabilities, and transformed large-scale payment data into tourism and market intelligence across Asia, Europe, and the U.S.
- Migrated SpendingPulse's near-real-time processing to Databricks Serverless, modernizing Spark and Delta Lake workflows to reduce operating costs and improve developer iteration.
- Redesigned the e-commerce adjustment step in a core SpendingPulse pipeline with an idempotent design, improving execution speed, maintainability, and downstream economist review.
- Built MCP servers for self-hosted infrastructure and supported cross-team on-call, triaging production incidents across unfamiliar dependencies and coordinating fixes with downstream teams.
- Built innovation-week tools adopted by the team for statistical synthetic-data generation and automated documentation auditing across Confluence and source repositories.

Software Engineer Trainee

Valeo R&D Center

December 2024 – April 2025

Prague, Czech Republic

- Built internal workflow applications with Google Apps Script, automating cross-department processes worth an estimated €77,000 in annual time and cost savings.

EDUCATION

Czech Technical University (ČVUT)

Bachelor of Science in Electrical Engineering and Computer Science

August 2022 – June 2025

Prague, Czech Republic

- **Machine Learning Research Thesis — CERN ATLAS**
- Developed a 1.2M-parameter Transformer classifier for rare Higgs boson signals in CERN ATLAS data, achieving an AUC of 0.804 and F1-score of 0.705. Built particle-specific preprocessing and regularization pipelines and interpreted predictions through self-attention aligned with physical observables.

PROJECTS

Assistant MK1 — Agent Operations Workbench

Next.js, TypeScript, WorkOS, Cloudflare Workers, D1, Durable Objects, LangGraph, Fly.io

May 2026 – Present

- Built a code-first agent workbench that makes durable runs, approvals, tool access, artifacts, and recovery inspectable outside the chat interface.
- Designed a tenant-safe control plane across Vercel, Cloudflare, and Fly, with server-derived identity, policy-enforced tool execution, audit trails, and durable run history.
- Implemented reusable agent packs, signed webhooks and schedules, failure alerts, and reconnect and retry workflows for safely operating long-running agents.

TECHNICAL SKILLS

Languages: Python, TypeScript, SQL, Kotlin, C#, C

Systems: Control-plane architecture, API boundary design, event-driven systems, durable job orchestration

Full-Stack: React, Next.js, .NET, Expo, Redux, Tailwind CSS, Uniwind, Redis, Postgres

Cloud & DevOps: AWS, Terraform, Cloudflare Workers, Docker, GitHub Actions, Tailscale, Fly.io, Railway

Data / ML Tools: Apache Spark, Databricks, PyTorch, Optuna, Pandas

CERTIFICATES

Databricks Certified Data Engineer Associate | *Databricks Cloud*

October 2024

Machine Learning Specialization | *Stanford Online – Andrew Ng*

September 2024